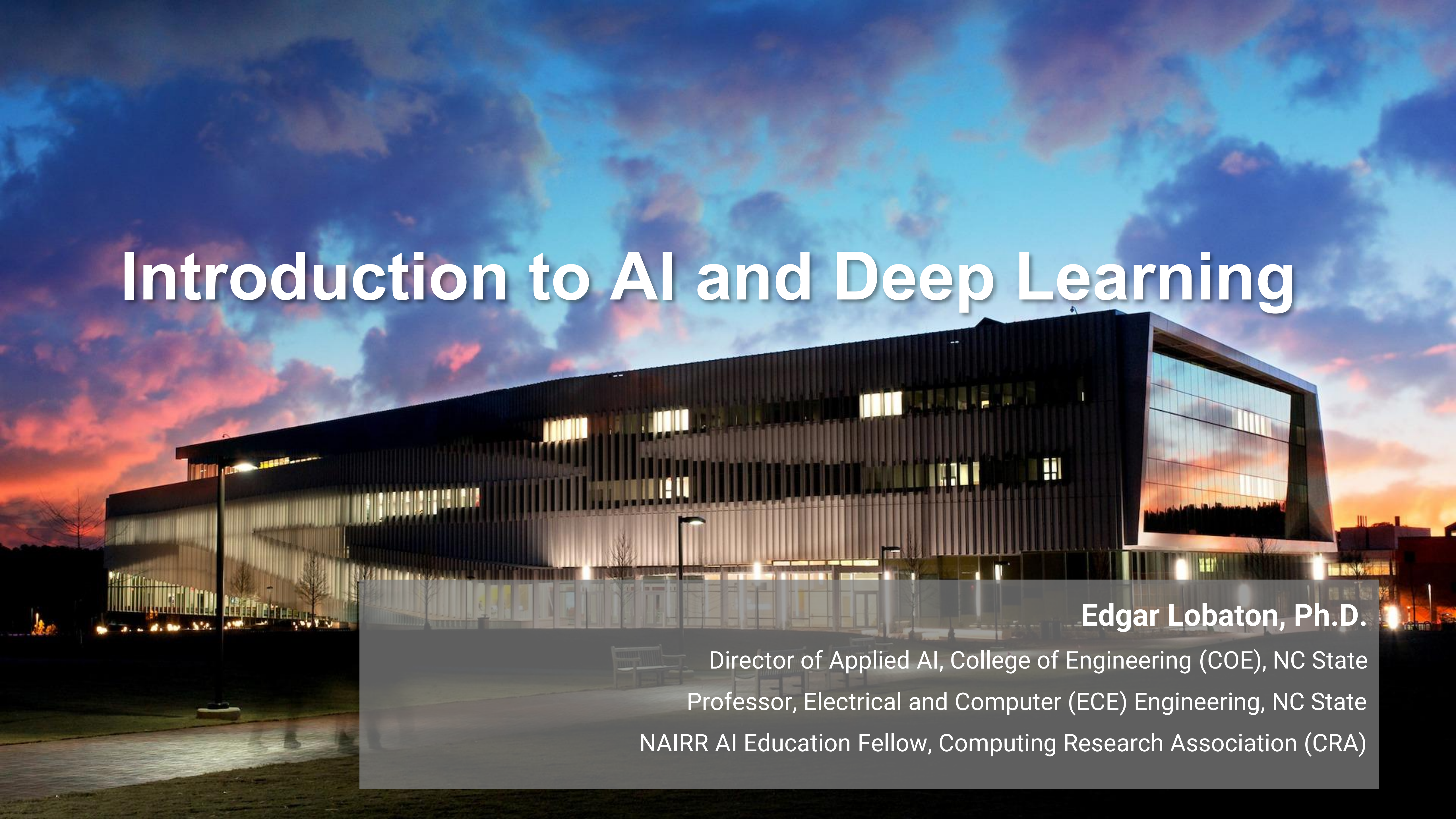

Introduction to AI and Deep Learning

Edgar Lobaton, Ph.D.

Director of Applied AI, College of Engineering (COE), NC State

Professor, Electrical and Computer (ECE) Engineering, NC State

NAIRR AI Education Fellow, Computing Research Association (CRA)

Course Overview

https://github.com/lobaton/DM2026-AI-and-Deep-Learning

What are AI and Machine Learning?

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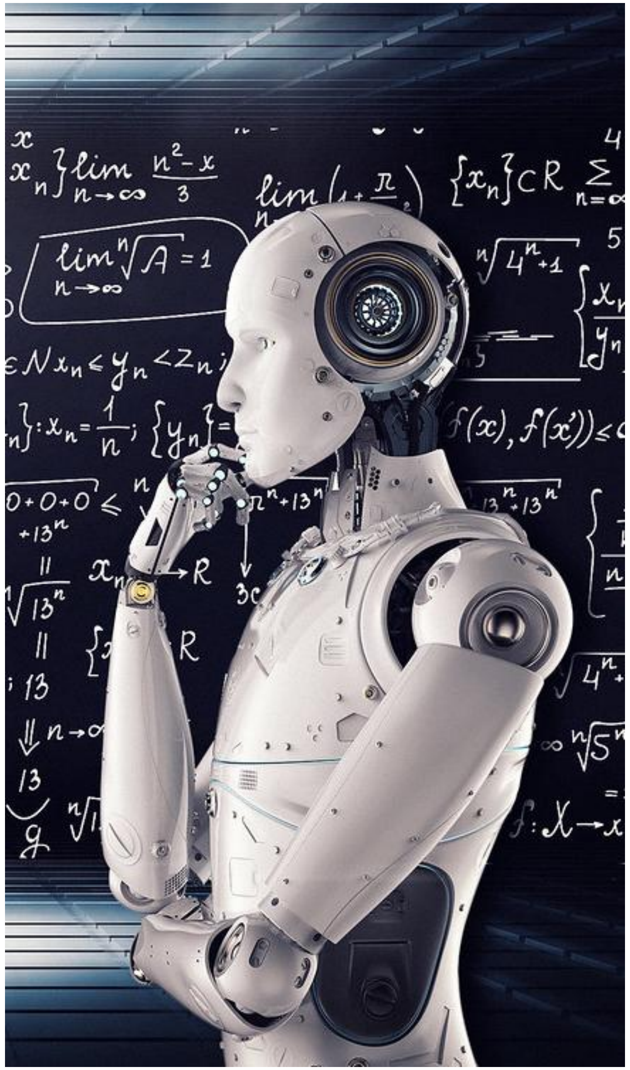
What is AI?

- AI allows computers to perform tasks that typically require human intelligence. AI systems can "learn" from information, recognize patterns, predict, make decisions and act on them.
- **Examples:** Generative AI (ChatGPT, Meshy AI), recommendation (Netflix, Amazon), and code assistants (Codex, Claude Code).
- A course on AI studies and designs intelligent agents. These agents can perceive their environment, reason about information, and take autonomous actions to achieve a specific goal.

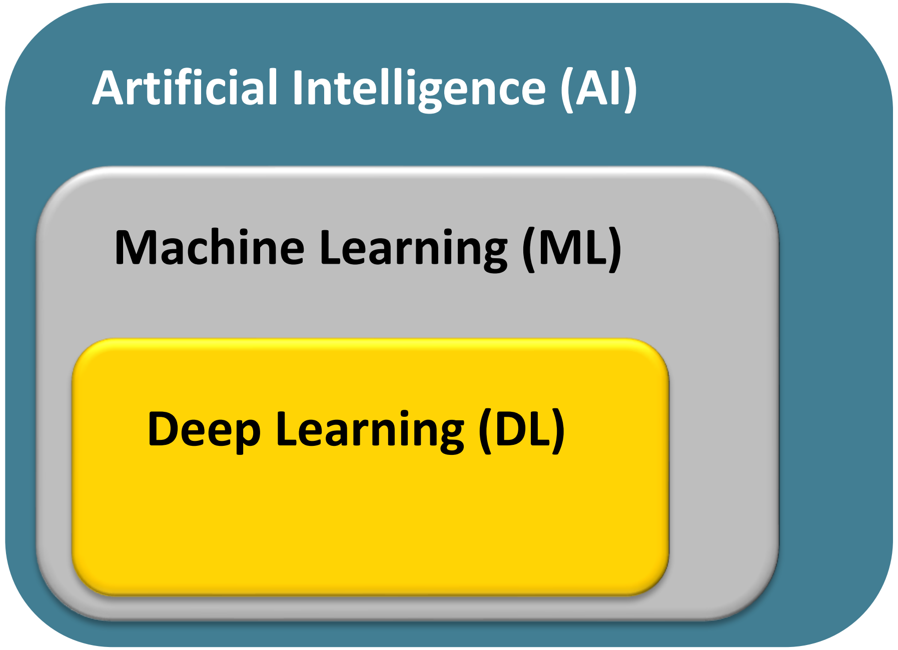


What is ML?

- **Machine Learning (ML)** is a subset of AI that focuses on algorithms that improve through experience. Instead of a human writing specific rules for every scenario, the programmer writes algorithms that allows the computer to find patterns in data.

- **Classical ML** requires feature engineering (i.e., humans tell the computer what parts of the data are important).

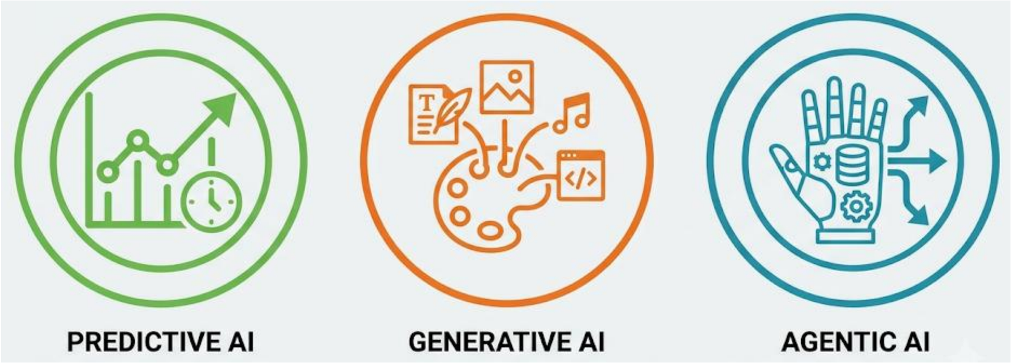
- **Deep Learning (DL)** is a subset of ML that discovers the features. Neural Networks are the foundation of DL.



Types of AI

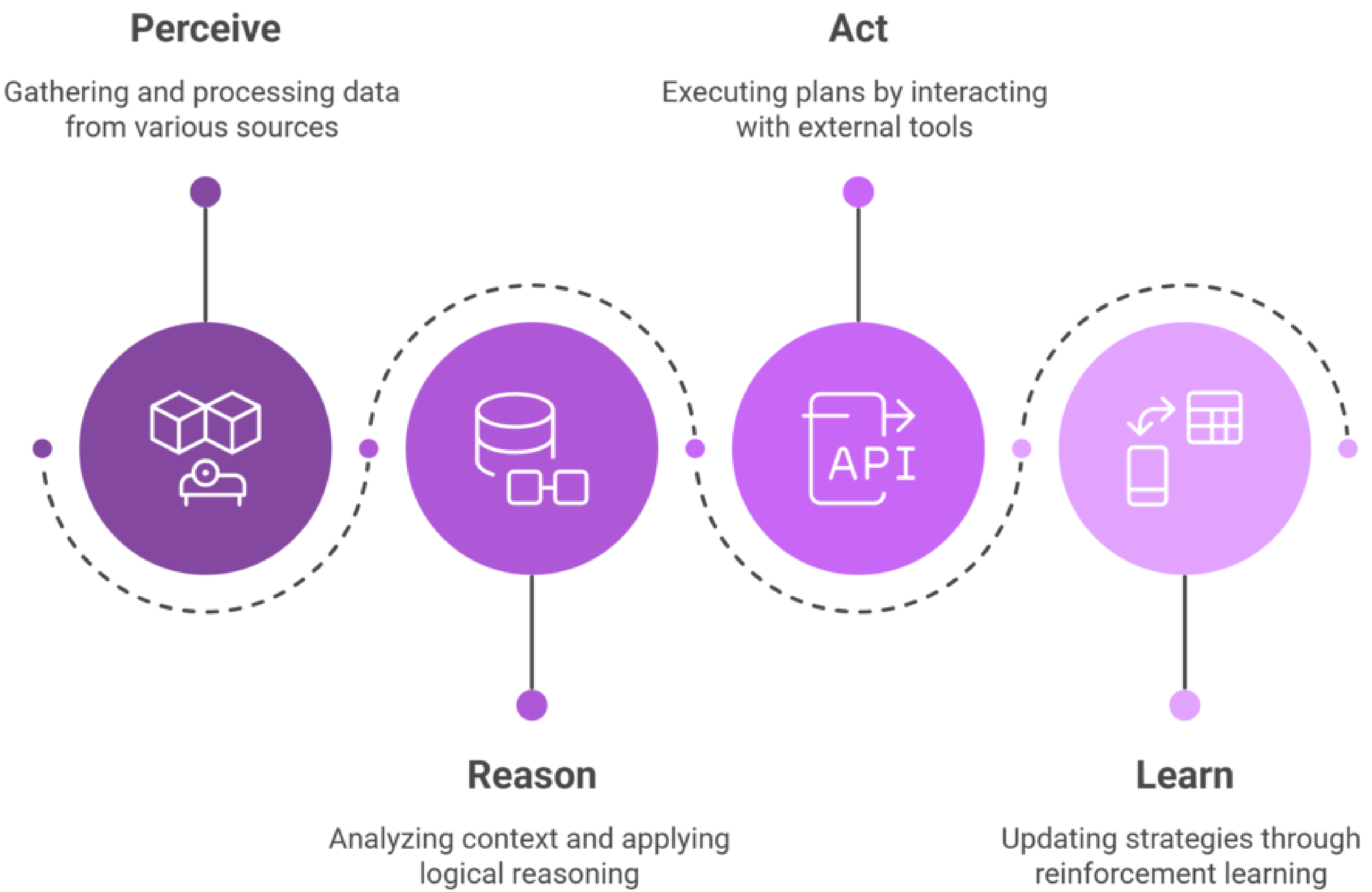
- **Predictive AI: What will happen?**
  - Produces a forecast, score or classification
  - Uses pattern recognition on historical data
  - Examples: Predictive maintenance / weather forecast

- **Generative AI: What can be created?**
  - Produces new content (text, images, code)
  - Uses sampling from a learned distribution
  - Examples: Generate a design / draft a report



- **Agentic AI: What should be done?**
  - Completes multi-step tasks
  - Uses reasoning and tools through multiple iterations often taking advantage of generative models.
  - Examples: An agent that runs a simulation, checks results and revises without a human in the loop.

Agentic AI Pipeline



* Figure from https://empathy-technologies.com/what-is-agentic-ai-the-complete-guide/

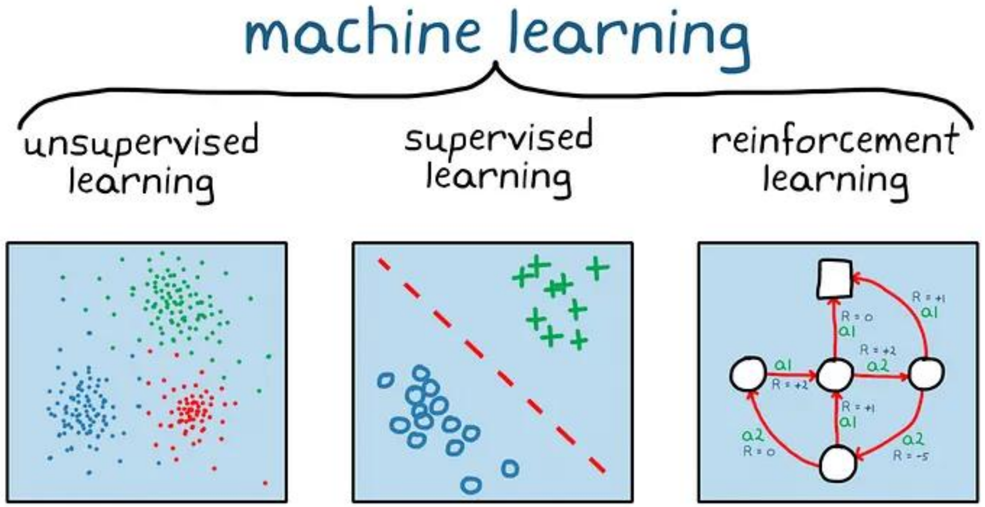
Types of AI - Scenarios

https://forms.gle/4bUMCWrYhGrao3zCA

Types of ML

- **Supervised Learning:**
  What is the output for this input?
  - Uses labeled images
  - Learns a mapping from input to output
  - Example: Crack detection for labeled images

- **Unsupervised Learning:**
  What pattern exists in this data?
  - Uses unlabeled examples
  - Finds groups, structure or distributions
  - Example: Clustering pump failures signatures from sensor logs



- **Reinforcement Learning:**
  What action maximizes reward over time?
  - Learns through trial-and-error
  - Takes actions, receives rewards, adjusts behavior
  - Example: A robotic arm learning to grasp an object

* Figure from https://www.mathworks.com/discovery/reinforcement-learning.html

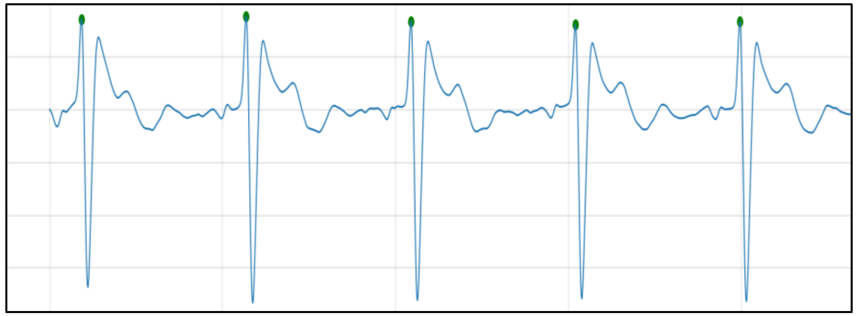
Types of ML - Scenarios

https://forms.gle/TaU3qWCbmXPi9ppg6

Supervised Learning

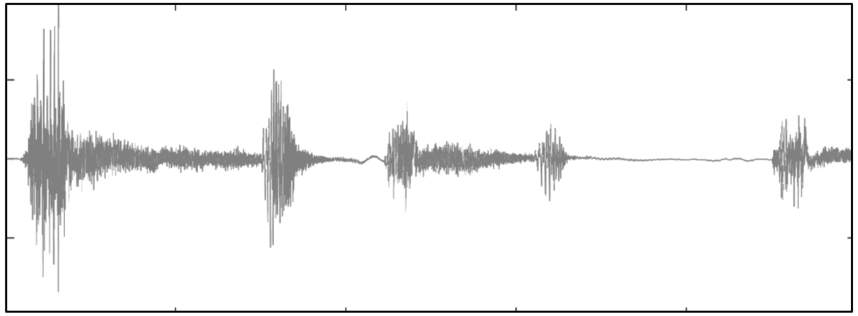
Given that data is provided in input/output pairs, then ML can help identify the mapping between them. That is, $\text{output} = f(\text{input})$.

Cat



90 bpm

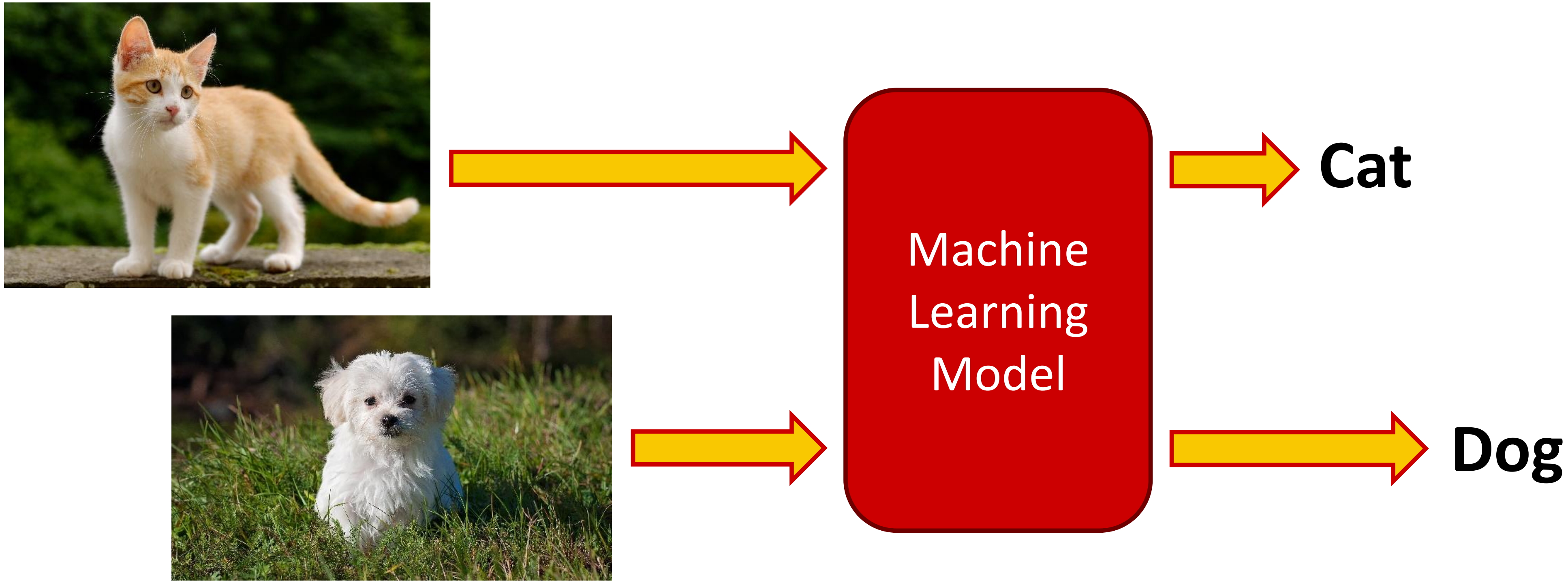
Dog



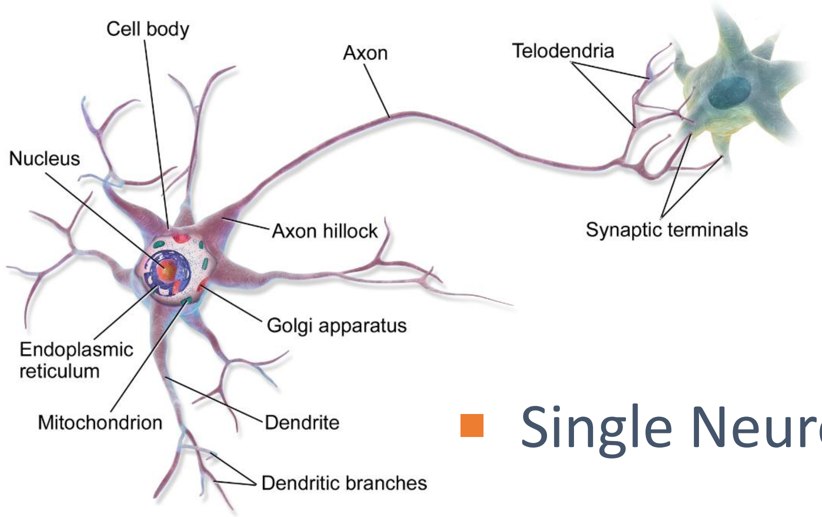
Stress

Supervised Learning

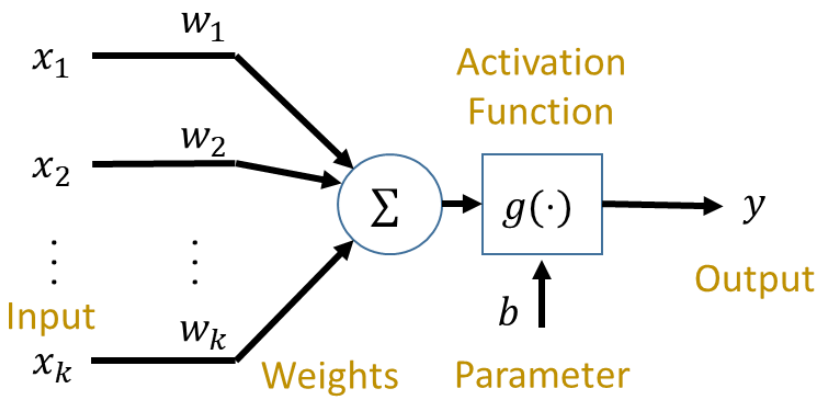
Think of a supervised learning model as a function or a black box (at least for now) that maps inputs (images) to outputs (labels in this case)

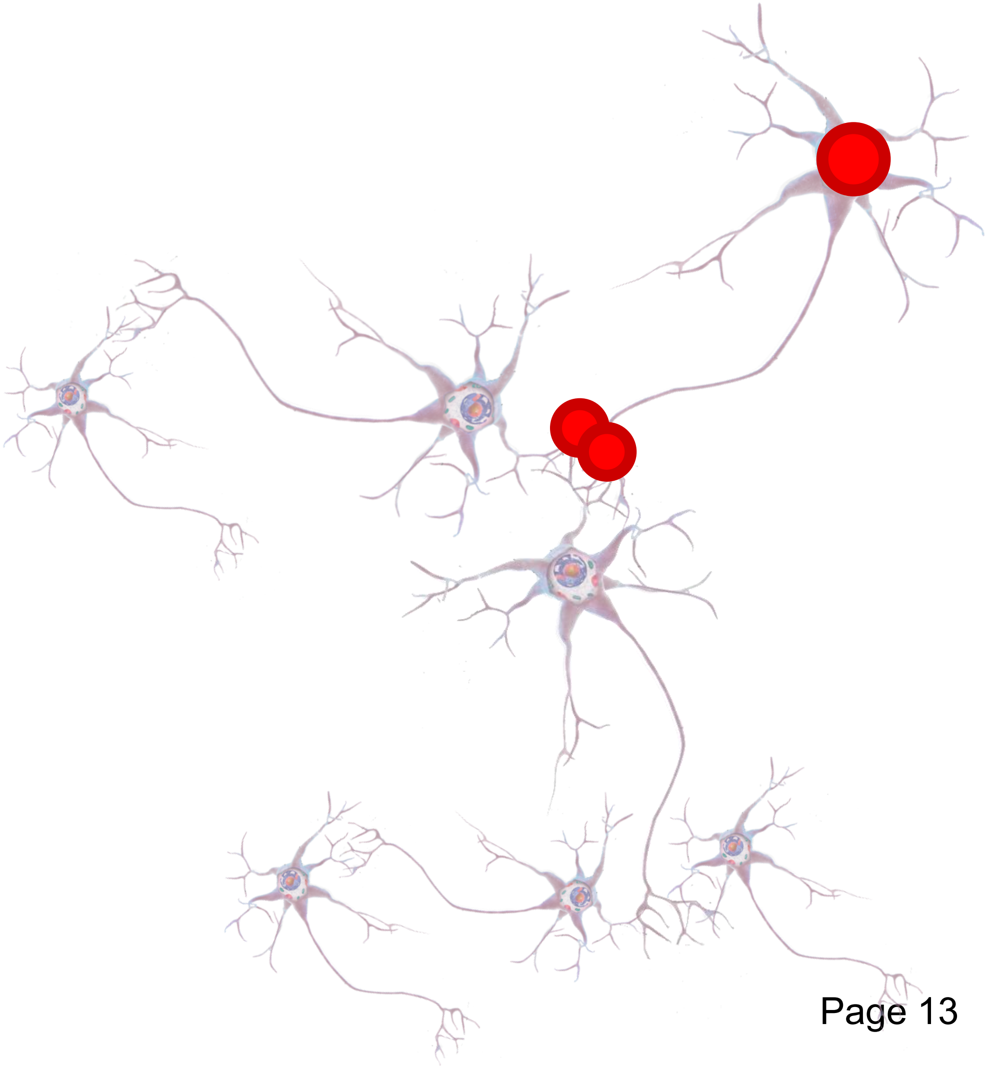


Artificial Neural Network (NN)



- Single Neuron

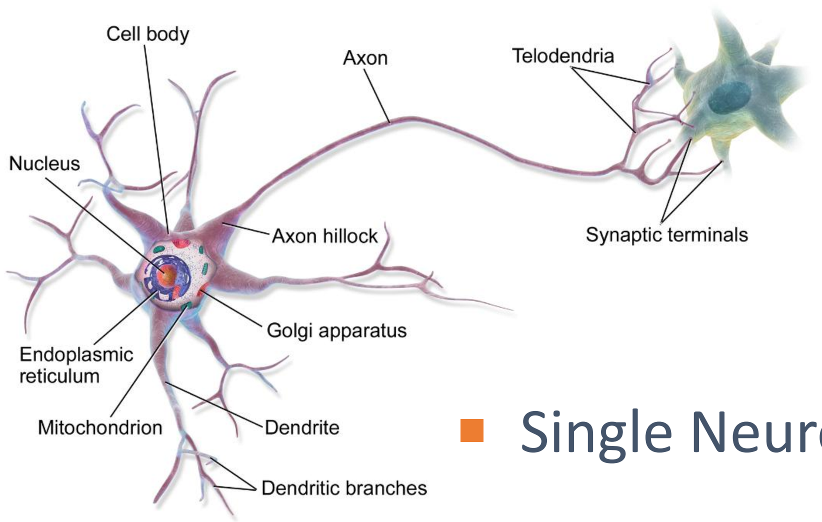




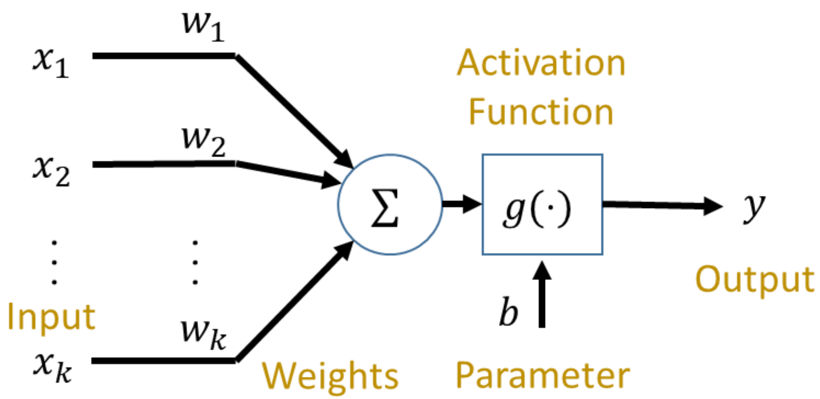
* Neurons figure modified from https://en.Wikipedia.org

NC STATE Engineering

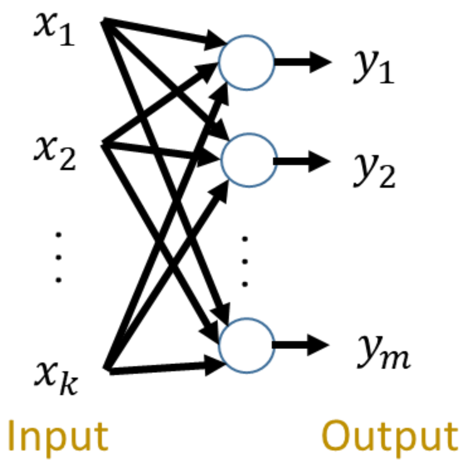
Artificial Neural Network (NN)



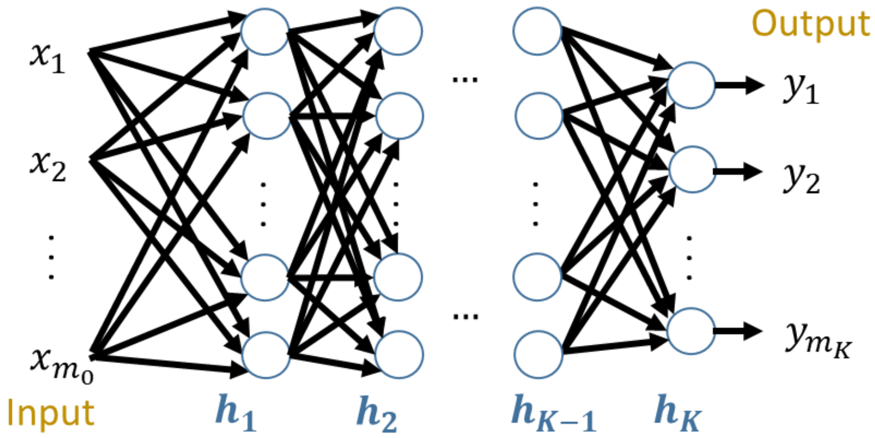
- Single Neuron



1-layer NN



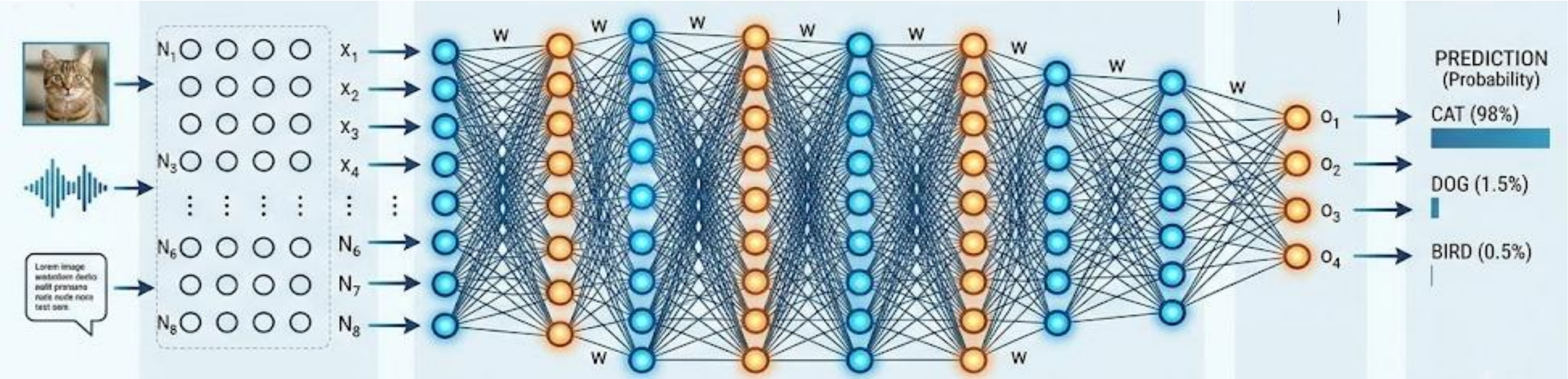
K-layer NN



* Neurons figure modified from https://en.Wikipedia.org

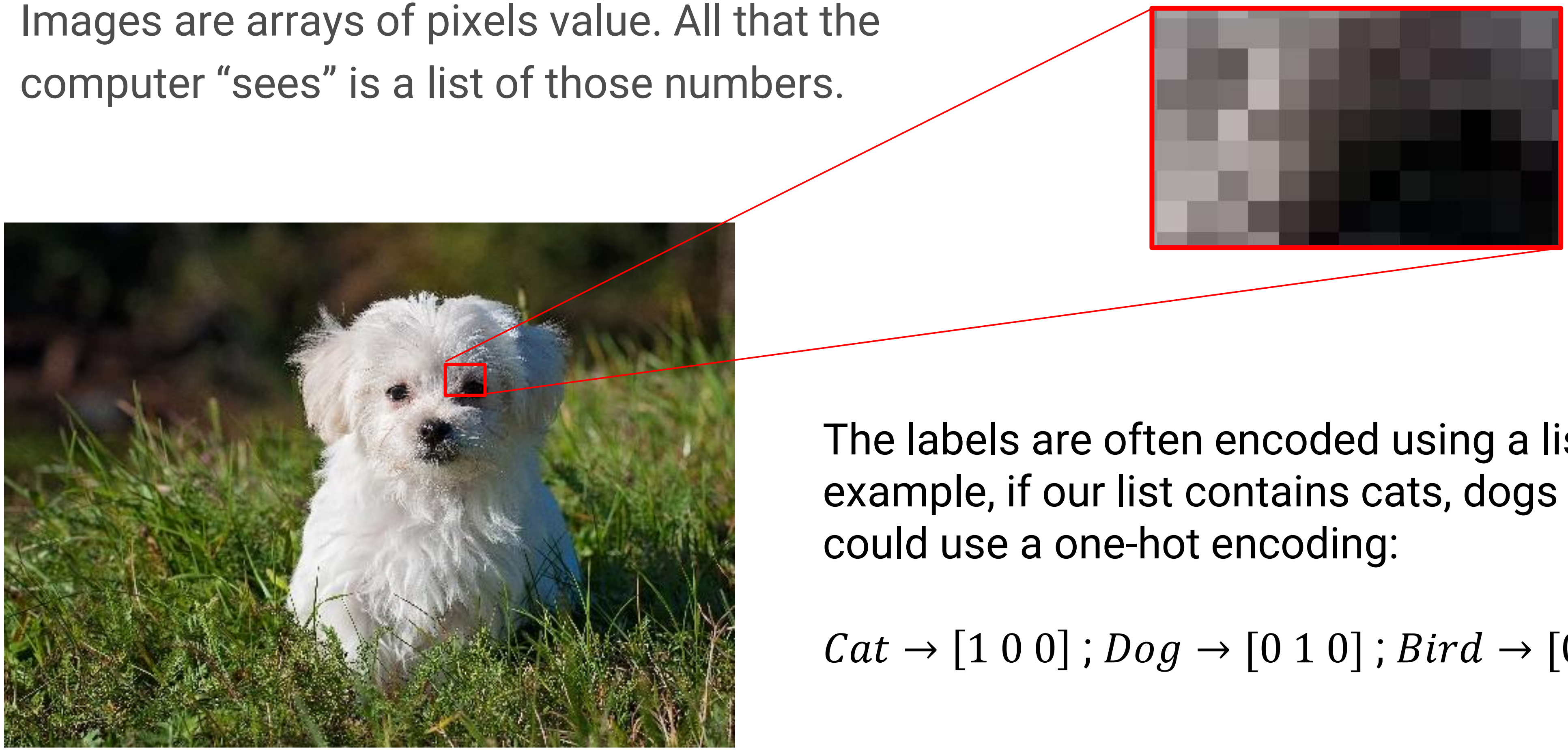
Deep Neural Network (DNN)

- A **DNN** is a NN with multiple layers and capable of learning features from raw data
- Traditional ML requires hand-craft features (e.g., computing statistical summaries); deep learning learns features directly from raw data, reducing that manual step
- Same core math as the 1980s backprop era — what changed is data availability, GPU compute, and architectural refinements (e.g., 2017 transformer architecture)



Understanding the Input and Output

Images are arrays of pixels value. All that the computer "sees" is a list of those numbers.



The labels are often encoded using a list as well. For example, if our list contains cats, dogs and birds, we could use a one-hot encoding:

$$Cat \rightarrow [1\ 0\ 0]\ ;\ Dog \rightarrow [0\ 1\ 0]\ ;\ Bird \rightarrow [0\ 0\ 1]$$

How Computers Learn

- Consider the training data
  [1 1] → 2
  [2 1] → 3
  [2 2] → 4
- What is the output for an input [2 1]?
- What is the output for an input [3 2]?

- Consider the training
  [1 1] → 3
  [2 1] → 5
  [2 2] → 6
- What is the output for an input [3 2]?
  Relationship 1: $[x\ y] \rightarrow\ 2x + y = 8$
  Relationship 2: $[x\ y] \rightarrow\ x^2 - x + y + 2 = 10$

We test an ML model by giving it data that it has not seen to ensure that it is capturing the underlying relationship. [**Generalization**]

The model needs to be properly specified given the amount of data provided. [**Overfitting / Underfitting**]

Exploring a Code Assistant

Let's test Colab